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The late Neoproterozoic Sierra de las Ánimas Magmatic Complex and Playa Hermosa Formation, southern Uruguay, revisited: Paleogeographic implications of new paleomagnetic and precise geochronologic data

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ABSTRACT

A new paleomagnetic study was carried out on 15 sites of the Sierra de las Ánimas complex, exposed in SE Uruguay. A paleomagnetic remanence was isolated at six sites, and this direction is likely to be pretectonic and primary. Mean directions from five of these sites are consistent with the SA2 paleomagnetic pole, which is based on a previous paleomagnetic study of the complex. A new paleomagnetic pole was computed for compiled results from the Sierra de las Ánimas complex (SAn, 12.2° S, 258.9° E, A95: 14.9°). A geochronological study based on U/Pb (SHRIMP) dating of magmatic zircons was also carried out on three samples of the Sierra de las Animas Complex, yielding a grand mean age of 578 ± 4 Ma. The new SAn paleomagnetic pole anchors the Early Ediacaran segment of the Apparent Polar Wander Path for the Río de la Plata craton. Consistency of the new pole with other Ediacaran poles from the Congo-São Francisco craton suggests a coherent Central Gondwana by this time. These data also support a wide Clymene Ocean between the conjoined Río de la Plata–Congo-São Francisco blocks and Western Gondwana, i.e., Amazonia–West Africa. Additional sampling (sixteen samples) was carried out on the basal levels of the glaciogenic Playa Hermosa Formation, exposed in the same area. These samples retain a paleomagnetic component consistent with the previously reported data, confirming the original pole position. U–Pb (SHRIMP) dating of detrital zircons reveal a bimodal distribution of Paleoproterozoic (around 2.1 Ga) and late Neoproterozoic ages. The youngest zircons are interpreted as pre-dating to penecontemporaneous with deposition and points to a ≤594 Ma age for deposition of the older levels of the Playa Hermosa Formation. A minimum age of deposition is set by the 578 ± 4 Ma age of the Sierra de las Ánimas magmatism. The low inclination of magnetization indicates that low-latitude (13.0° +9.5°/−5.5°), Gaskiers-aged glacial deposits are preserved atop the Rio de la Plata craton.

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1. Introduction

The late Proterozoic chronology of the fragmentation of Rodinia and the assembly of Gondwana is poorly constrained (e.g. Li et al., 2008; Murphy et al., 2013). Reliable paleomagnetic data on accurately dated rocks are essential to reconstruct the paleogeographic evolution of the different cratonic blocks. The relevant time period should encompass the entire period over which Rodinia disintegrated and Gondwana formed, from around 800 up to 500 Ma. The Río de la Plata craton (RP, Fig. 1) is a relatively small crustal block with Paleoproterozoic (and very minor Archean) basement, that forms the southernmost cratonic unit of South America (Dalla Salda et al., 1988; Cingolani and Dalla Salda, 2000; Cordani et al., 2000; Rapela et al., 2007, 2011; Bossi and Cingolani, 2009). The paleogeographic position of the Rio de la Plata craton over the 800–500 Ma interval is poorly known. Some workers favor a close proximity of

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Fig. 1. Main geotectonic units of the Río de la Plata craton and location of the study area in southern Uruguay. SYSZ: Sarandí del Yi shear zone. Modified from Rapalini et al. (2013).

the Rio de la Plata craton to Amazonia, and therefore to Eastern Laurentia (e.g. Weil et al., 1998; Meert, 2001; Collins and Pisarevsky, 2005; Li et al., 2008; Gaucher et al., 2011), though no conclusive geologic evidence supports that reconstruction. In contrast, several geological studies (e.g. Campos Neto, 2000; Tohver et al., 2012) have shown that the Rio de la Plata was surrounded by several Neoproterozoic oceanic domains that were closed by the end of that era, and might have never been part of Rodinia (Kröner and Cordani, 2003; Tohver et al., 2006; Rapela et al., 2007).

A review of the available Neoproterozoic paleomagnetic data from the Rio de la Plata craton suggests that this block was not connected to Eastern Laurentia in the Ediacaran (Rapalini et al., 2013). These authors defined an apparent polar wander path (APWP) for the Río de la Plata craton from around 600 to 500 Ma based on eleven paleomagnetic poles. This APWP is marred by known uncertainties: some of these poles should be regarded as mean virtual geomagnetic poles due to low number of sites or samples, and some of the poles are based on remagnetizations, the age of which is inferred from the pole position, while other poles show poor confidence ovals and/or age uncertainties. Despite the progress made in the last decade, a more robust APWP for the Rio de la Plata craton is clearly needed.

The Sierra de las Ánimas Magmatic Complex, exposed in SE Uruguay, provided the first late Neoproterozoic to early Paleozoic paleomagnetic results for the Río de La Plata craton (Sánchez Bettucci and Rapalini, 2002). Two paleomagnetic poles (SA1 and SA2) were calculated from this complex with assigned ages to these poles of ca. 520 and 550 Ma, respectively. This interpretation indicates either long-lived magmatism (over 30 Ma) or two discrete magmatic events, one in the Ediacaran (SA2) and one in the Cambrian (SA1). However, in the absence of precise geochronological determinations for these rocks, these age assignments are weakly constrained and inferential. More recently, Oyhantçabal et al. (2007) reported a well-behaved, $^{40}Ar/^{39}Ar$ plateau age of 579 ± 2 Ma for hornblende from the Pan de Azúcar syenite, which would imply a significant revision to the age of the SA2 pole, assuming that the hornblende does not record protracted cooling of the body. This geochronological uncertainty aside, it must be stated that both the SA1 and SA2 poles are based on a relatively small number of sites and consequently show large uncertainties in their positions. A third mean virtual geomagnetic pole reported by Sánchez Bettucci and Rapalini (2002) derived from study of the glaciogenic sedimentary rocks of the Playa Hermosa Formation. This pole, computed from only 6 samples, was interpreted as primary in origin and dating from the inferred ca. 600 Ma age of sedimentation.

In order to construct a more robust apparent polar wander path for the Río de la Plata craton, we conducted a new paleomagnetic study on the Sierra de las Animas complex and the Playa Hermosa Formation. This paleomagnetic study was complemented by geochronologic determinations by U–Pb (SHRIMP)

of magmatic zircons of the igneous complex and detrital zircons from the sedimentary rocks. The expanded paleomagnetic study and combination with reliable geochronological data for both studied units contribute to a more reliable Ediacaran APWP for the Rio de la Plata craton and constrain its paleogeography during the critical transition between the Rodinia and Gondwana supercontinents.

2. Geology and sampling

The Sierra de Las Ánimas complex is an association of intrusive, volcanic, and sedimentary rocks generated during an extensional stage within the evolution of the Neoproterozoic Brasiliano-Pan African orogeny (Sánchez Bettucci and Rapalini, 2002). The studied area is situated in southeastern Uruguay (Fig. 1), and presents a complex stratigraphic sequence. The Sierra de las Ánimas complex is a bimodal igneous suite of both shallow intrusive bodies and volcanic, pyroclastic deposits. Trachytes are the most abundant and geographically widespread. Extrusive units comprise pyroclastic rhyolites and minor basalts. According to Oyhantçabal et al. (2007), shallow intrusive bodies like the Pan de Azúcar syenite and associated microsyenitic dykes comprise the shallow magmatic plumbing of the complex. These same workers reported an $^{40}Ar/^{39}Ar$ cooling age of 579 ± 2 Ma for the Pan de Azúcar syenite, considered as part of the Sierra de Las Animas Complex.

The Playa Hermosa Formation (Masquelin and Sánchez Bettucci, 1993) is a volcano-sedimentary succession exposed mainly along the coastline close to the town of Piriápolis (Fig. 2). It has been interpreted as a glacially influenced clastic sequence, at least in its lower part, that makes up the infilling of the Playa Verde basin (see Pazos et al., 2011 and references therein, for a detailed description of this formation). The upper part of this succession shows a large number of volcanic intercalations, either as synsedimentary lavas or sills or as dykes. This volcanism has been ascribed to the Sierra de las Ánimas complex. A preliminary paleomagnetic study on the basal levels of the Playa Hermosa Formation was reported by Sánchez Bettucci and Rapalini (2002). Based on only six sample directions, these authors obtained a mean virtual geomagnetic pole that falls close to the 595 Ma Campo Alegre pole (D'Agrella-Filho and Pacca, 1988), which was interpreted as a likely age of deposition of ca. 600 Ma for the basal levels of the Playa Hermosa Formation. The position of this preliminary paleomagnetic pole suggested that the glacial event associated to the lower levels of this formation may have occurred at low to intermediate latitudes.

We collected one hundred and fifteen oriented samples with a portable drill at fifteen sites distributed in different magmatic lithologies as basalt, trachyte and micro-syenite of the Sierra de las Ánimas complex. This new sampling was done with the aim of improving the statistical parameters of poles SA1 and SA2 published by Sánchez Bettucci and Rapalini (2002). Samples were also collected for geochronologic studies at three sites that yielded positive paleomagnetic results in the original study. They were the Pan de Azucar syenite (sites SA-1, SA-2, SA-3 and SA-8, see also Oyhantçabal et al., 2007) and a micro-syenite (SA-20), both contributing to pole SA2 (interpreted as of ca 550 Ma) and the El Tambo Rhyolite (SA-12) contributing to pole SA1 (interpreted as of ca. 520 Ma).

Sixteen new paleomagnetic samples were collected at the basal levels of the Playa Hermosa Formation. They were approximately distributed into four sites, although no proper sites were defined since all samples were collected over c.4.5 m of stratigraphy. One sample was also collected from these levels for U–Pb (SHRIMP) dating on detrital zircons in order to determine source and maximum depositional ages for the Playa Hermosa Formation.

3. Laboratory procedures

Paleomagnetic samples were submitted to standard stepwise demagnetization techniques. Alternating field (AF) demagnetization proved to be more efficient in those sites that presented well-defined magnetic components. Specimens were submitted to steps of 3, 6, 9, 12, 15, 20, 25, 30, 40, 50 and 60 mT in most cases, with demagnetization steps of up to 100 mT used for some samples. Remanence directions were measured in a fully automated 2G DC cryogenic magnetometer. AF demagnetization was performed in a static three-axis degausser attached to the magnetometer. Thermal demagnetization was accomplished in a dual-camera ASC non-magnetic oven, with internal magnetic fields under 10 nT. Bulk susceptibility was monitored with a Bartington MS-2 susceptometer and anisotropy of magnetic susceptibility with a MFK-1A kappabridge (AGICO). All laboratory procedures were done at the Daniel A. Valencio paleomagnetic laboratory (IGEBA, Universidad de Buenos Aires). Magnetic components were defined by means of principal component analysis (Kirschvink, 1980) after visual inspection of demagnetization diagrams (Zijderveld plots) for each specimen. Maximum angular deviations up to 15° were accepted, although over 85% of all determined components showed MAD <10°. Mean directions for each site were computed following standard Fisherian statistics (Fisher, 1953). In just one site combination of isolated vectorial components and remagnetization circles (McFadden and McElhinny, 1988) was employed. In the case of the Playa Hermosa Formation no mean site directions were computed as each sample corresponded to a slightly different stratigraphic level.

Zircon crystals were separated by crushing and sieving of samples, followed by Wilfley table and heavy liquid separation. Grains were then handpicked using a binocular microscope, and then mounted in epoxy resin for polishing prior to scanning electron microscope and cathodoluminescence imaging (Fig. 3). The epoxy mounts were cleaned and gold-coated to have a uniform electrical conductivity for SHRIMP analysis. The zircon standard used was BR266 zircon (559 Ma, 903 ppm U). The isotopic composition of the minerals was determined using the SHRIMP II housed at Curtin University. Prior to spot analysis, rastering of the ion beam was carried out for 120–150 s to remove the gold coating and reduce the common Pb contaminant within the gold coating. A primary ion beam of 2.5–3 nA with a diameter of ~25 μm was focused onto the polished surface. The common Pb correction was carried out using the measured amount of ^{204}Pb. Isotope data were reduced using the SQUID2 software (Ludwig, 2009). Data were plotted on concordia diagrams using the Isoplot 3 software (Ludwig, 2008), with 2σ error ellipses on the concordia plots and all ages reported at the 2σ confidence level in the text.

4. Geochronologic results

4.1. The Sierra de las Animas Complex

Three samples were collected from the sites studied by Sánchez Bettucci and Rapalini (2002) in order to constrain the age of the rocks and the observed paleomagnetic remanences. Samples SdAn1 and SdAn3 are from sites (site SA-20 and sites SA-1, SA-2, and SA-3, respectively) representing the older SA2 direction, originally considered to be ca. 550 Ma. Sample SdAn1 is a microsyenitic dyke intruding the upper levels of the Playa Hermosa Formation along the beach near Piriápolis. All ages discussed here are $^{238}U–^{206}Pb$ ages; U–Pb analytical data and location details for all samples are provided in Supplementary Table S1. Zircons from sample SdAn1 are characterized by Th/U ratios >0.5 and as high as 2, consistent with a magmatic origin for these grains. SHRIMP spot analyses from cores and rims failed to distinguish age zoning, consistent

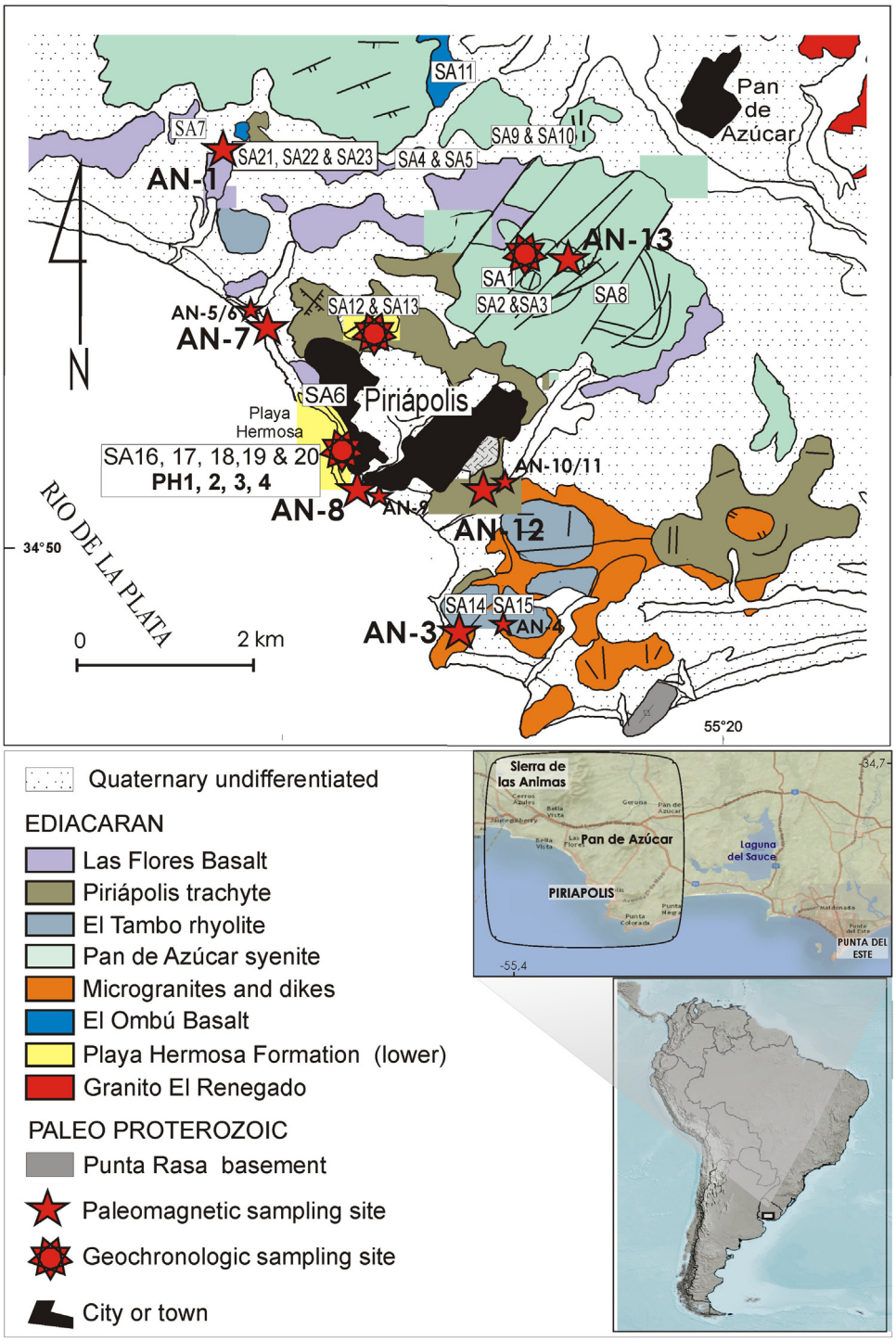


Fig. 2. Geologic map of the study area with location of both paleomagnetic and geochronologic sampling sites. Location of sites from the previous study by Sánchez Bettucci and Rapalini (2002) are indicated and labelled SA. Large stars show sites with positive paleomagnetic results in the new study.

with the igneous origin ascribed to these grains. Inheritance from Paleoproterozoic (ca. 2.0 Ga, two grains) and late Mesoproterozoic (ca. 1.0 Ga, one grain) sources is observed, as well as a single grain affected by substantial Pb loss. The dominant population for this sample is Ediacaran in age, with a "concordia" age of 574.5 ± 8.1 Ma calculated from six grains (Fig. 4A).

Sample SdAn3 is a medium-grained syenite of the Sierra de las Animas complex taken from a quarry. Fewer zircons were obtained from this sample, although the Th/U ratios of the analyzed grains are similar to, if not higher than those from sample SdAn2, ranging from 0.8 to >4, with an average Th/U ratio of ca. 2.5. Again, no significant age zonations are evident. No inheritance from older sources is observed from this sample, but this probably reflects the relatively small number of zircon grains and smaller analyzed population. Slight discordance and some scatter of results precludes calculation of a mean-weighted U–Pb age for this sample, so an upper intercept age of 587.4 ± 7.3 Ma represents the best approximation of the age of this sample (Fig. 4B).

Sample SdAn2 is a flow-banded rhyolite of the El Tambo Formation, taken from site SA-12 of Sánchez Bettucci and Rapalini (2002).

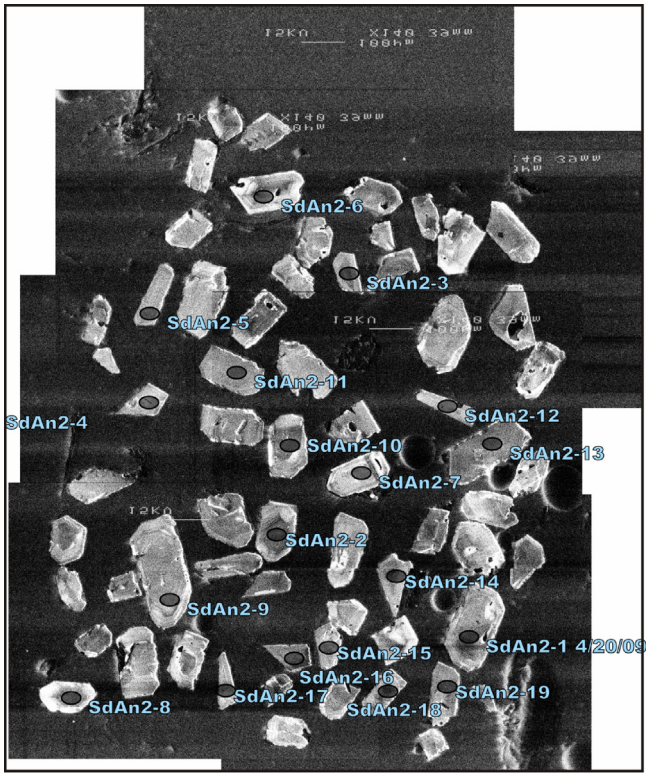


Fig. 3. Cathodo-luminiscence image of zircons analyzed from sample SdAn-2 (El Tambo Rhyolite, paleomagnetic site SA-12 from Sánchez Bettucci and Rapalini, 2002).

This site was characterized by a different paleomagnetic remanence, the basis of paleomagnetic pole SA1, and was tentatively assigned a ca. 520 Ma age by those authors. Zircons from this sample produced a very well defined age on Concordia of 581.8 ± 3.4 Ma (11/18 grains, Fig. 4C), with Th/U ratios consistent with those observed for the other two samples. No inherited component is observed in this sample, and very few grains show evidence of Pb loss. This result indicates that the El Tambo rhyolite is the extrusive equivalent of the syenitic intrusive rocks. Taking the three age determinations together, we calculate a grand mean weighted U–Pb age of 578.0 ± 4.3 (2σ) Ma for the Sierra de las Animas complex. The implications of these results for the paleomagnetic data are discussed below.

4.2. The Playa Hermosa Formation

Forty-one detrital zircons from a sample collected at the basal levels of this formation were analyzed by U–Pb SHRIMP. Considering only concordant analyses with common Pb contamination of <1.5%, the ages of these grains define a bimodal population (Fig. 5), with ca. 60% of zircons with ages of 2050–2200 Ma, typical of the Paleoproterozoic basement of the Río de la Plata craton (Rapela et al., 2007). The second, younger group represents 40% of the population, and the median age of this younger population is ca. 615 Ma (+6/−10 Ma, 96.5% confidence level). Using the "youngest detrital zircon" function of Isoplot, the youngest Monte Carlo result is 593 (+11/−18) (Fig. 5b, inset), in close agreement with the youngest single analytical result from a zircon with a $^{206}Pb/^{238}U$ age of 594 ± 16 Ma (2σ). The source for this younger population is likely to be the products of extensional magmatism in the adjacent Dom Feliciano Belt. Noting that the Sierra de las Ánimas is intrusive into the Playa Hermosa Formation, the age of deposition is constrained to a period between ca. 594 Ma and the 578.0 ± 4.3 Ma intrusion, consistent with the 590–600 Ma age imputed to these basal deposits by Sánchez Bettucci and Rapalini (2002).

5. Paleomagnetic results

5.1. The Sierra de las Animas Complex

Only six out of sixteen sites from this complex provided reliable and consistent paleomagnetic results after detailed demagnetization. The remaining ten sites generally showed unstable magnetic behavior during demagnetization or very poor directional consistency of remanence. In all cases alternating field (AF) demagnetization proved more efficient than thermal cleaning to isolate and determine the magnetic components. Fig. 6 illustrates some examples of samples with positive demagnetization behavior. In all cases a well-defined magnetic component is erased by fields between 15 and 60 mT. In some cases (Fig. 6a and b) some 10–20% of the original remanence may remain after 60 mT. In all cases a linear trend towards the origin is observed. This was interpreted as the characteristic remanence of these rocks. Five out of these six sites (AN-1, 3, 7, 8 and 13) presented mean directions consistent with those corresponding to the six sites from which pole SA2 (Sánchez Bettucci and Rapalini, 2002) was originally calculated. These eleven sites' mean directions are shown in Table 1 and plotted in Fig. 6d. Site AN-12 presented a well-defined mean direction that is not consistent with the remaining sites and was ruled out. The Sierra de las Ánimas Complex outcrops are not affected by significant deformation, only minor tilting has affected some areas. Therefore, just two sites have been corrected for tilting (Table 1). Fig. 6d and e shows a better clustering of mean site characteristic directions after bedding correction, suggesting a pre-tilting magnetization. Application of the McElhinny (1964) statistical fold test is inconclusive (Fisherian statistical parameter k jumps from 12 to 20), but the direction–correction test of Enkin (2003) produces a positive result with best grouping at 98.6% of bedding correction. These results strongly suggest in favor of a pre-tilting magnetization.

A virtual geomagnetic pole was computed from the average characteristic remanence of each site (Table 1). These were averaged with those mean site directions presented by Sánchez Bettucci and Rapalini (2002). The overall mean is the new paleomagnetic pole for the Sierra de las Ánimas complex (SAn: 12.2° S, 258.9° E, A95: 14.9°). Considering the ages obtained from paleomagnetic sites for this pole, in particular SA20, an age of 578 Ma can be applied confidently to such paleomagnetic pole.

Surprisingly not a single site yielded a remanence direction compatible with the group of directions determined by Sánchez Bettucci and Rapalini (2002) from which paleomagnetic pole SA1 was computed.

5.2. The Playa Hermosa Formation

Sixteen oriented cores were collected at the basal levels of the Playa Hermosa Formation on siltstones to fine-grain sandstones of glacially related origin. These samples can be roughly grouped into four sites and were aimed at improving the original results published by Sánchez Bettucci and Rapalini (2002).

The anisotropy of magnetic susceptibility (AMS) was measured for all samples (23 specimens) prior to demagnetization, and the data were analyzed using the standard statistical approach of Jelinek (1978, 1981). Results are presented in Fig. 7. The bulk susceptibility of these rocks was homogeneous with a mean value of 3×10^{-4} SI, suggesting that the paramagnetic fraction is likely dominant (Tarling and Hrouda, 1993). Fig. 7c and d shows that

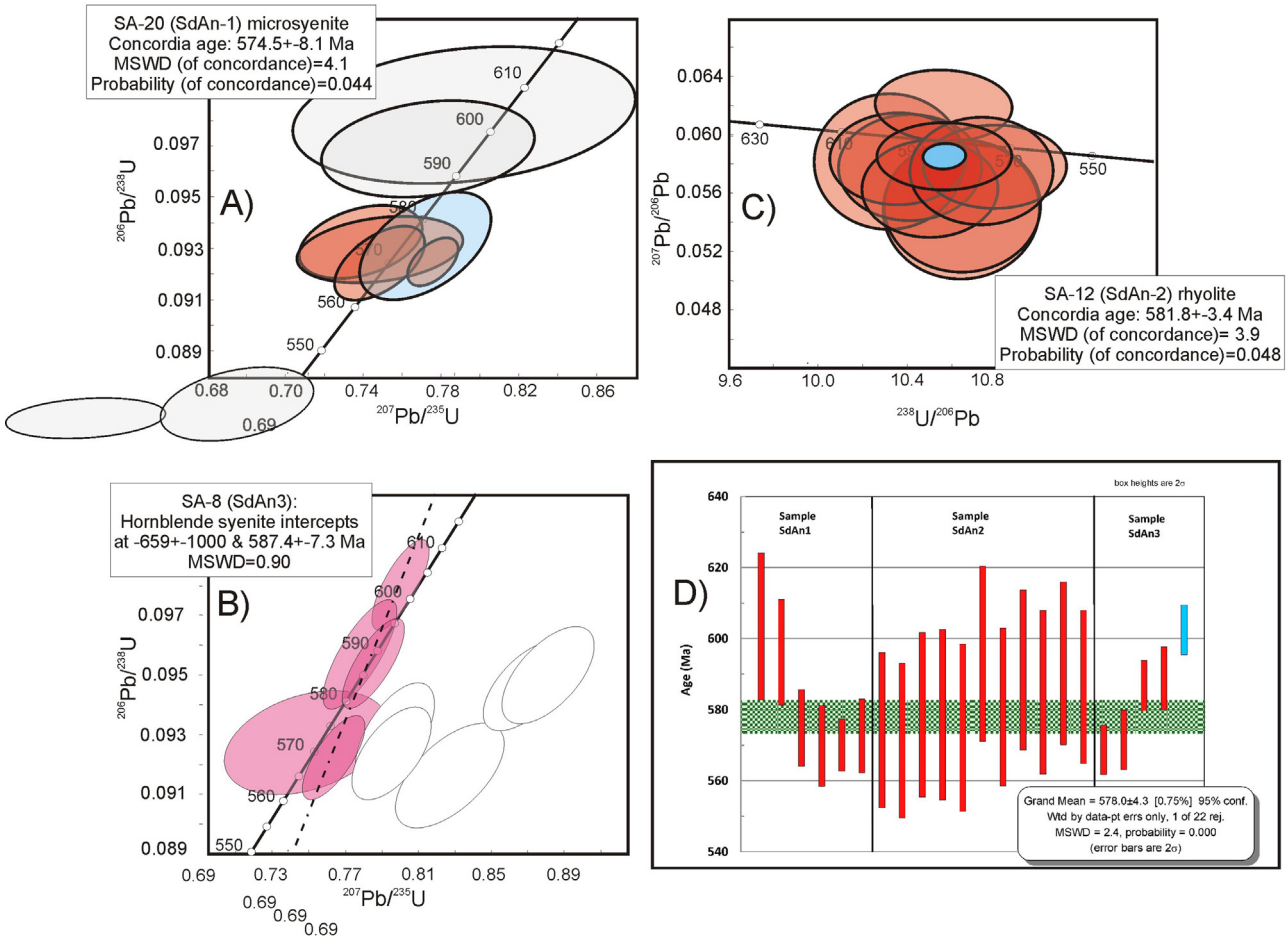


Fig. 4. (A–C) Concordia plots for magmatic zircons from three samples of the Sierra de las Ánimas complex. (D) Grand mean for the U–Pb ages from the three samples of the Sierra de las Ánimas complex. Experimental data is presented in Supplementary Table S1. See discussion in the text.

the anisotropy degree is low (under 5%), and the positive shape parameter (T) indicates an oblate fabric. This is consistent with the absence of pervasive deformation and metamorphic overprint in these rocks (Pazos et al., 2011). The distribution of principal axes of susceptibility is presented both in situ (Fig. 7a) and after restoring the sedimentary bedding to a horizontal plane (Fig. 7b). The dominant oblate fabrics of these sedimentary rocks is clearly observed in Fig. 7a and b, in which k_{min} ($k3$) axes appear well grouped while k_{max} ($k1$) and k_{int} ($k2$) define a girdle. Intringuingly, the magnetic fabric of the Playa Hermosa rocks is marked by $k3$ axes oriented in a horizontal ESE direction, while the magnetic foliation plane defined by the $k1$–$k2$ axes is NNE–SSW and vertical. Since the exposed succession of these sediments is homoclinal, dipping moderately (40–45°) towards the NW, restoration of the beds to the paleohorizontal

Table 1
Mean site paleomagnetic data for the Sierra de las Ánimas complex. Dec: declination, Inc: inclination; α95: 95% confidence angle around mean direction; k: precision parameter (Fisher, 1953), *Dec** and *Inc** are the declination and inclination values after restoration to paleohorizontal, n (N) indicate the number of samples (sites) used to compute the mean. In italics (sites SA), data from Sánchez Bettucci and Rapalini (2002). ** Mean site direction obtained from two magnetic components defined by principal component analysis and three remagnetization circles, applying the algorithm by McFadden and McElhinny (1988). B: basaltic flow, Sy: syenite, SyD: microsyenitic dyke, TBr: trachytic breccia. North (south) latitudes are positive (negative). All are east longitudes.

Site	Rock type	n (N)	*Dec* (°)	*Inc* (°)	α95 (°)	Strike/dip	*Dec** (°)	*Inc** (°)	α95 (°)	VGP		
										Lat (°)	Long (°)	A95 (°)
AN-1	B	10	165.8	−64.9	13.0	0/0	165.8	−64.9	13.0	7.3	295.0	
AN-3	TBr	5**	302.1	54.6	19.7	0/0	302.1	54.6	19.7	1.6	260.8	
AN-7	TBr	11	120.5	−48.5	13.7	0/0	120.5	−48.5	13.7	4.7	255.9	
AN-8	SyD	7	182.3	−62.5	15.7	120°/42°	66.5	−68.9	15.7	−40.7	257.0	
AN-13	Sy	3	135.0	−55.5	15.1	0/0	135.0	−55.5	15.1	7.6	269.5	
SA-4	*B*	*2*	*91.6*	*−67.3,*	–	0/0	*91.6*	*−67.3*	–	*−25.1*	*259.6*	
SA-5	*B*	*7*	*107.6*	*−55.2*	*7.9*	0/0	*107.6*	*−55.2*	*7.9*	*−7.6*	*253.4*	
SA-7	*B*	*4*	*48.5*	*−60.2*	*15.3*	0/0	*48.5*	*−60.2*	*15.3*	*−51.8*	*239.0*	
SA-8	*Sy*	*2*	*291.8*	*60.7*	–	0/0	*291.8*	*60.7*	–	*−8.8*	*260.2*	
SA-11	*B*	*3*	*295.4*	*36.5*	*21.0*	0/0	*295.4*	*36.5,*	*21.0*	*7.6*	*246.0*	
SA-20	*SyD*	*9*	*16.0*	*−54.2*	*8.2*	*232°/39°*	*79.2*	*−57.8*	*8.2*	*−28.5*	*243.8*	
Mean		(11)	112.5	−63.4	14.0 (k:12)		108.0	−60.6	10.4 (k:20)	−12.2	258.9	14.9

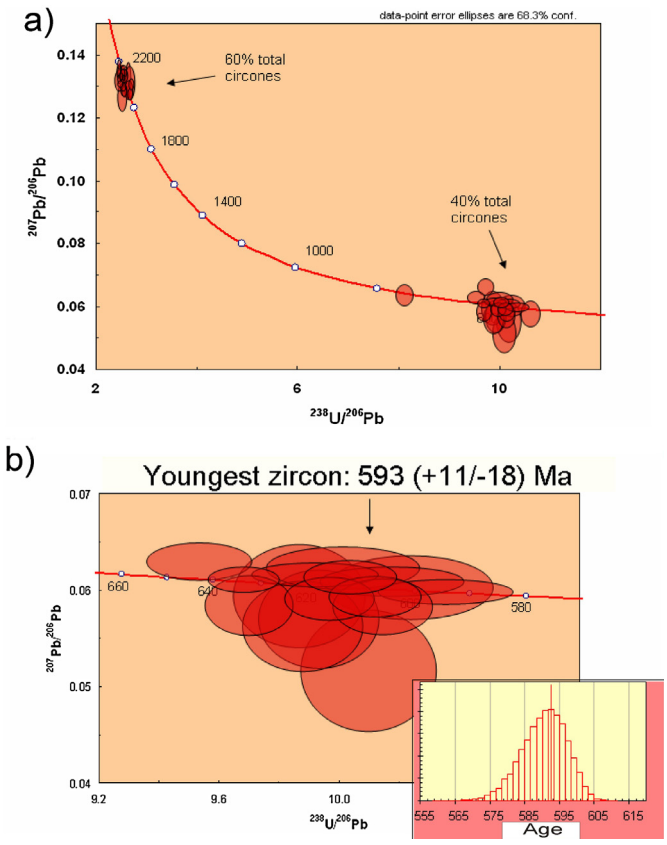


Fig. 5. (a) Concordia plot for detrital zircons of the Playa Hermosa Formation. Note the bimodal distribution; (b) concordia plot of the youngest group of zircons. The inset histogram gives the simulated age distribution of the youngest single zircon grain. Experimental data is presented in Supplementary Table S1.

(Fig. 7b) produces a peculiar distribution of AMS axes, in which there seems to be no relation between the bedding and the foliation planes. Although no remanence anisotropy has been investigated in these samples, the bulk susceptibility values suggest that the magnetic fabric is probably due to phyllosilicates population, which are the modally dominant Fe-bearing minerals.

The geometrical disposition of the AMS axes resembles that found in several clastic sedimentary rocks and unconsolidated sediments and attributed to macroscopically imperceptible deformation due to tectonic processes during or immediately after deposition (e.g. Sagnotti and Meloni, 1993; Mattei et al., 1997; Parés et al., 1999; Soto et al., 2009; Fazzito et al., 2011). This magnetic "cryptofabric" has more frequently been determined in compressional settings. However, Cifelli et al. (2005) have described similar magnetofabrics in extensional basins. In all cases the orientation of AMS axes (in particular *k*max) is governed by the orientation of the regional or local strain ellipsoid. This phenomenon has been generally described in Neogene sediments or sedimentary rocks. A possible explanation for the magnetic fabrics found in the Playa Hermosa Formation is that these rocks record a tectonic "cryptofabric" coeval with their deposition or early diagenesis. Unfortunately, no unambiguous field evidence on the age of the folding/tilting of the Playa Hermosa sediments is available. One favored hypothesis suggests that it is related to extensional tectonic disturbance associated to the emplacement of the Sierra de las Ánimas magmatism. If so, the age of its magnetic fabric might postdate deposition no more than 10–20 Ma, consistent with the interpretation presented above.

According to the results published by Sánchez Bettucci and Rapalini (2002), samples from the Playa Hermosa Formation presented more coherent demagnetization behavior when submitted to AF rather than thermal cleaning. Maximum applied alternating fields reached 120 mT. Although several samples showed unstable magnetic behavior, a characteristic magnetic component could be isolated in 7 samples from two sites (Fig. 8a and b), in fields between 12 and 60–80 mT. These samples generally showed a low-coercivity (viscous?) large overprint with random direction that could be easily erased at fields under 12 mT. As previously found by Sánchez Bettucci and Rapalini (2002) a small high coercivity component (>60 mT) was also found in some samples (Fig. 8b), although with low directional consistency. The characteristic remanence points down and southwards (Table 2). Fig. 8c and d shows the newly determined sample directions together with those previously published by Sánchez Bettucci and Rapalini (2002). Bedding attitudes of the sampled strata shows slight differences between sites. However, the homoclinal nature of the exposed succession renders any tilt test as indeterminate. Response to isothermal remanence acquisition tests (Sánchez Bettucci and Rapalini, 2002) and petrographic observations suggest that the remanence in the sediments of the Playa Hermosa Formation is carried by detrital titanomagnetite.

A new sample-based mean direction for the Playa Hermosa Formation was computed based on the tilt-corrected remanence (*Dec*: 207.6°, *Inc*: 27.3°, *n*: 13, *k*: 12.1, α95: 12.4°). A mean virtual geomagnetic pole for these rocks was computed by averaging the virtual geomagnetic poles obtained from each sample: 58.8° S, 183.1° E, A95: 12.1°.

Table 2
Sample characteristic remanence directions for the Playa Hermosa Formation. MAD: maximum angular deviation (Kirschvink, 1980). References as in Table 1.

Sample	Site	Dec (°)	Inc (°)	MAD (°)	Strike/dip (°)	Dec (°)	Inc (°)	VGP	
								Long (°)	Lat (°)
P2	PH1	142.0	18.0	7.0	212/46	165.0	55.0	27.2	−77.7
P3A	PH1	179.0	14.0	6.0	212/46	196.8	27.6	165.6	−64.7
P3B	PH1	178.0	14.0	6.0	212/46	196.1	28.3	164.8	−65.5
P4	PH1	167.0	21.0	5.0	212/46	193.9	40.8	174.7	−73.3
P11B	PH4	178.0	17.0	7.0	215/40	192.4	21.2	152.9	−63.5
P11A	PH4	180.0	8.0	12.0	215/40	187.7	12.2	140.6	−60.9
P12	PH4	190.1	11.0	10.0	215/40	197.4	9.0	156.6	−55.6
16 3a	SA16	214.3	30.6	6.8	210/41	234.0	20.0	201.7	−35.2
16 6a	SA16	194.3	53.6	4.7	210/41	245.5	45.5	225.9	−34.2
16 7a	SA16	215.2	18.2	8.3	210/41	225.9	10.3	190.4	−38.3
17 1a	SA17	203.0	12.3	13.5	203/42	211.3	9.1	175.5	−48.1
17 6	SA17	186.2	32.1	8.4	203/42	215.1	34.0	196.3	−55.0
17 7a	SA17	204.5	34.8	14.7	203/42	228.9	24.2	200.4	−40.7
Mean (n = 13)		186.6	22.9			207.6	27.3	183.1	−58.8
		α95: 11.8°				α95: 12.4°		A95: 12.1°	

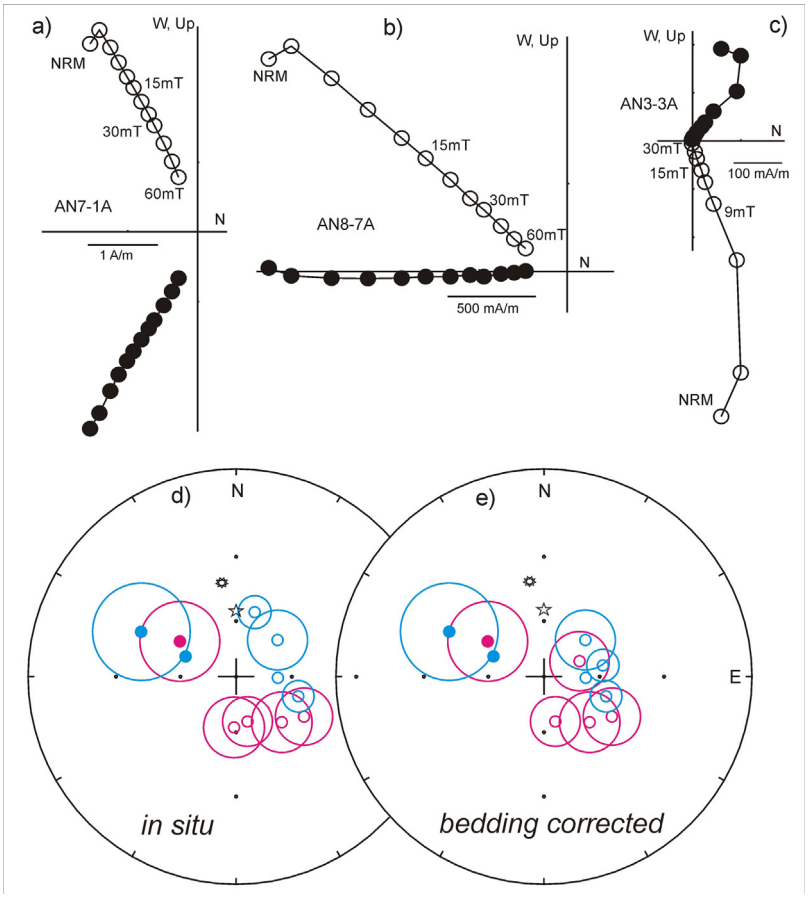


Fig. 6. (a–c) Representative Zijderveld plots of samples from different sites of the Sierra de las Ánimas complex submitted to stepwise demagnetization. Full (open) symbols represent end-vector projections on the horizontal (vertical) plane; (d) in situ mean site characteristic remanence directions for the Sierra de las Ánimas complex including our results (in red) and those from Sánchez Bettucci and Rapalini (2002) (in blue); (e) in situ mean site characteristic remanence directions for the Sierra de las Ánimas complex including our results (in red) and those from Sánchez Bettucci and Rapalini (2002) after bedding correction. Data is presented in Table 1. Full (open) symbols correspond to downward (upward) directions. Five-pointed star: geomagnetic dipole direction; eight-pointed star: present-day geomagnetic field direction at the study locality. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

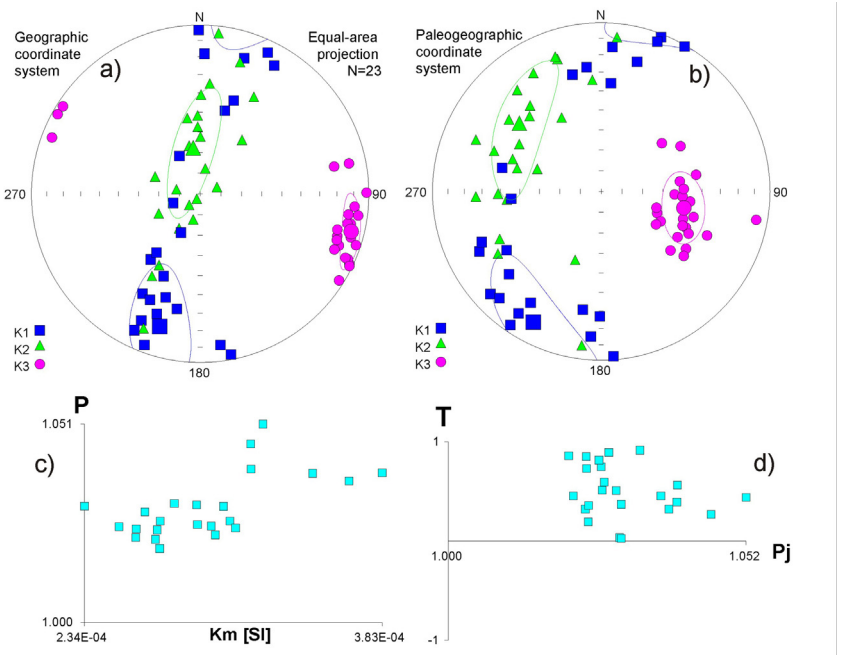


Fig. 7. (a) In situ distribution of anisotropy of magnetic susceptibility axes for the Playa Hermosa Formation samples; (b) in situ distribution of anisotropy of magnetic susceptibility axes for the Playa Hermosa Formation samples after bedding correction; (c) anisotropy degree (Pj) vs mean susceptibility (Km) plot and (d) shape parameter (*T*) vs anisotropy degree (Pj) plot for the Playa Hermosa Formation samples.

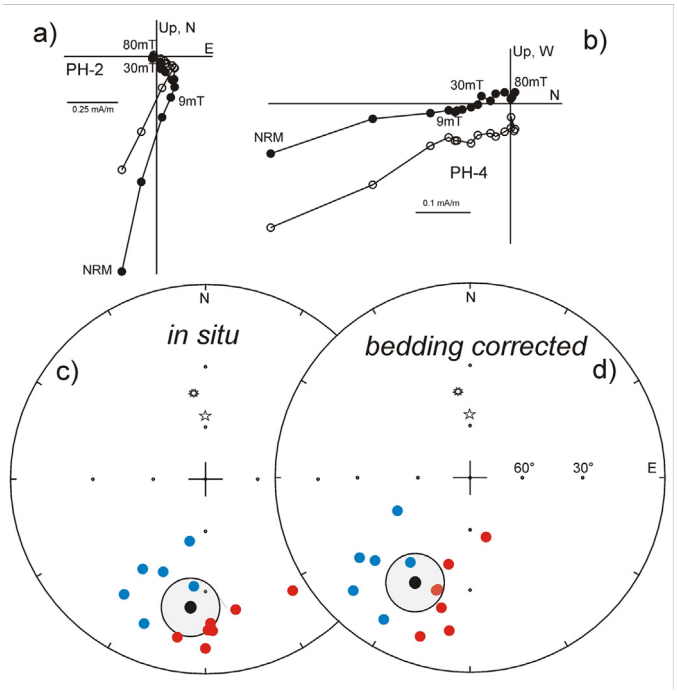


Fig. 8. (a, b) Representative Zijderveld plots of samples from the Playa Hermosa Formation submitted to stepwise demagnetization; (c) in situ sample characteristic remanence directions for the Playa Hermosa Formation including our results (in red) and those from Sánchez Bettucci and Rapalini (2002) (in blue); (d) sample characteristic remanence directions for the Playa Hermosa Formation including our results (in red) and those from Sánchez Bettucci and Rapalini (2002) (in blue) after bedding correction. Overall mean is presented as a larger full circle with its α95. Data is presented in Table 2. Symbols as in Fig. 6. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

6. Discussion

6.1. Implications for the Rio de la Plata APWP and Gondwana assembly

Originally, Sánchez Bettucci and Rapalini (2002) proposed that the Sierra de las Animas magmatism extended for at least 30 Ma between ca. 550 and 520 Ma. The paleomagnetic basis for this interpretation lies with the two significantly different characteristic remanence directions observed at different sites in the complex: a moderately inclined, northeast and downward direction (SA1), and a dual polarity East and down (or West and up) with moderate to high inclinations (SA2). Lithological correlations using antiquated geochronological data were consistent with two magmatic episodes at ca. 550 Ma for SA2 and ca. 520 Ma for SA1, whose corresponding paleomagnetic poles presented large confidence ovals (Sánchez Bettucci and Rapalini, 2002). These workers also reported a small number of paleomagnetic results for the Playa Hermosa Formation, which was assigned a ca. 600 Ma age.

Our new results allow us to revise these results substantially, notably by rejecting the primary nature of the SA1 pole, and significantly constraining the age of the primary SAn pole (new results compiled with the SA2 pole and recalculated) from the Sierra de las Ánimas complex and the PH pole from the Playa Hermosa Formation (PH). With regard to the first point, we found no samples preserving the remanent direction underpinning the SA1 pole position. Furthermore, new geochronological data from the El Tambo Rhyolite indicate a ca. 578 Ma age for these rocks in conflict with the interpreted Cambrian (ca. 520 Ma) age for the magnetization. The newly obtained age strongly suggests that the Sierra de Las Ánimas Magmatic Complex formed during a relatively short time during the Early Ediacaran, and that pole SA1 is a likely remagnetization of Cambrian age.

The new SAn and PH poles are presented in Fig. 9 and Table 3 together with other poles or VGPs for the Rio de la Plata craton, the Congo-São Francisco craton and other Gondwana blocks (see Rapalini et al., 2013), in south African coordinates following the

Table 3
Selected Ediacaran to Cambrian paleomagnetic poles for the Rio de la Plata, Congo-São Francisco, West Africa, Laurentia and Baltica illustrated in Figs. 9–11. Poles have been rotated into African coordinates according to Reeves et al. (2004) and Trindade et al. (2006).

Geologic unit	PP	Lat	Long	A95 (dp/dm)	Age (Ma)	Method	Ref.
Río de la Plata craton							
Cerro Victoria F. (remag)	CVc	28.8°	13.2°	11°/16°	500?	APWP	Rapalini and Sanchez Bettucci (2008)
Sierra de Animas 1 (remag)	SA1	22.9°	8.7°	20°/27°	520?	K/Ar, APWP	Sánchez Bettucci and Rapalini (2002)
Cerro Negro F.	SBc	17.9°	354.4°	10°/13°	520?	Strat. & APWP	Rapalini et al. (2013)
Polanco F. (remag)	P	15.4°	359.3°	14°/17°	520?	APWP	Rapalini and Sanchez Bettucci (2008)
Olavarría F.	SBd	4.0°	322.8°	6°/9°	535?	APWP	Rapalini et al. (2013)
Los Barrientos	LB	−47.0°	312.9°	11°/14°	570?	APWP	Rapalini (2006)
Sierra de Animas Complex	SAn	−50.7°	311.1°	16°/21°	578	U–Pb, Ar–Ar	This paper
Cerro Largo F.	SBe	−66.9°	257.3°	8°/14°	580?	APWP	Rapalini et al. (2013)
Villa Mónica F.	SBf	−76.2°	159.5°	6°/11°	590?	APWP	Rapalini et al. (2013)
Playa Hermosa F.	PH	−74.1°	180.9°	9°/16°	594	U–Pb & Strat	This paper
Campo Alegre F.	CA	−80.5°	68.5°	10°	595	U/Pb	D'Agrella-Filho and Pacca (1988)
Congo-São Francisco							
Nola dykes	ND	−61.8°	304.8°	5°/11°	571	Ar–Ar	Moloto-A-Kenguemba et al. (2008)
West Africa							
Adma Diorite	AD	32.5°	344.7°	16°	613	U–Pb	Morel (1981)
Laurentia							
Sept-Iles	SI	−20°	321°	5°/9°	565	U–Pb	Tanczyk et al. (1987)
Catoctin Volcanics	CV	42°	297°	17°/17°	572	U–Pb	Meert et al. (1994)
Callander Complex	CC	46°	301°	6°/6°	577	K–Ar	Symons and Chiasson (1991)
Long Range dykes	LR	19°	355°	18°/25°	615	U–Pb	Murthy et al. (1992)
Baltica							
Baltica Mean	BM	−30°	118°	7°/12°	555	U–Pb	Meert et al. (2007)
Alno Carbonatite	AC	63°	101°	12°/14°	584	Ar–Ar	Meert et al. (2007)
Fen Complex	FC	57°	151°	7°/11°	583	Ar–Ar	Meert et al. (1998)

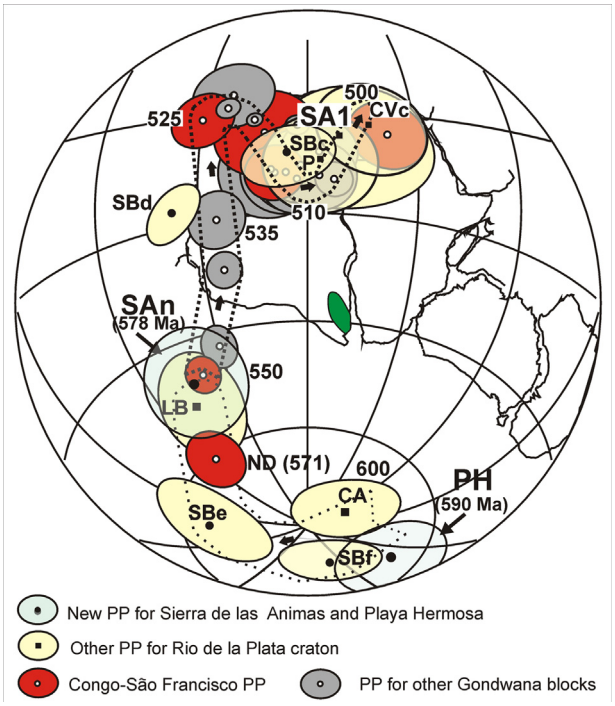


Fig. 9. Distribution of paleomagnetic poles for the Río de La Plata and other Gondwana cratons for the Ediacaran–Cambrian and the position of the newly calculated Sierra de Las Animas (SAn) and Playa Hermosa (PH) poles. Numbers indicate most likely ages of paleomagnetic poles. See Table 3 for labels and data for the poles. Those from the Rio de la Plata craton are rotated into African coordinates according to parameters presented by Trindade et al. (2006): 43.3° N, 329.9° E, 58.8°.

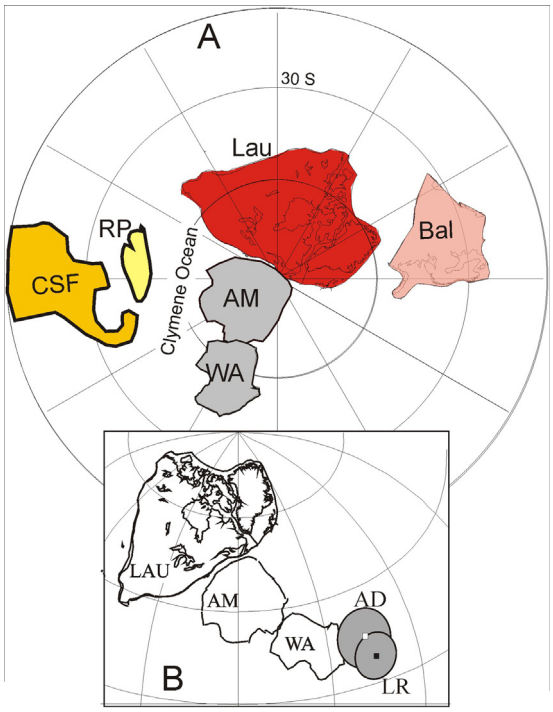


Fig. 10. (A) Paleogeographic reconstruction of Laurentia (Lau), Baltica (Bal), Amazonia (AM), West Africa (WA), Rio de la Plata (RP) and Congo-São Francisco (C-SF) cratons for ca. 575 Ma. RP and C-SF are positioned according to the new Sierra de las Ánimas paleomagnetic pole (SAn), Lau according to the Callander Complex (CC) pole, Bal following Cawood and Pisarevsky (2006). AM and WA are positioned following the paleomagnetically controlled reconstruction for ca. 615 Ma with respect to Lau (Sánchez Bettucci and Rapalini, 2002), by overlapping the coeval LR (Long Range dykes, Laurentia) and AD (Adma Diorite, West Africa) poles. See the inset figure (B). Note the existence of a large oceanic domain between Rio de la Plata and Amazonia, known as the Clymene Ocean. See discussion in the text.

"tight reconstruction" of Gondwana by Trindade et al. (2006), based upon Reeves et al. (2004). The new poles broadly confirm the shape of the APWP for this block (Rapalini, 2006; Rapalini et al., 2013), although the ages are modified. The SAn pole is assigned an age of 578 ± 4 Ma, slightly displaced from the original SA2 pole, becoming more consistent with the Los Barrientos pole (LB) from the Río de La Plata craton, the 547 Ma Sinyai dolerite pole (Meert and Van der Voo, 1996) from Congo and the Mirbat Sandstones pole (Kempf et al., 2000) from Arabia; while it moves away from the contemporaneous, 571 Ma pole from the Nola dykes (Moloto-A-Kenguemba et al., 2008).

The age discrepancies between these different poles prevent us from ruling out small magnitude movement between these cratons after ca. 578 Ma. However, the clustering of these poles, or possibly, their arrangement along a single APW path strongly argues in favor of a coherent "Central Gondwana" by that time *sensu* Sánchez Bettucci and Rapalini (2002) and Tohver et al. (2006). Refining the magnitude of possible displacements with respect to the assembly of the Congo-São Francisco and Río de La Plata cratons will need further paleomagnetic data from both cratons.

A comparison of the resultant APWP for the Rio de la Plata–Congo-São Francisco cratons with the position of the Amazon–West Africa cratons in Ediacaran times can help resolve the controversy over the Clymene Ocean (Trindade et al., 2006; Tohver et al., 2010, 2012; Cordani et al., 2013; Tohver and Trindade, 2014). The dispute concerns the timing of the final assembly of the cratonic blocks of South America by either 600 Ma (Cordani et al., 2013) or not until the middle Cambrian (Tohver et al., 2012). A simple comparison of the position of the ca. 600 Ma Adma diorite pole (Morel, 1981) of West Africa with the ≤594 Ma PH pole and ca. 578 Ma SAn pole within a reconstructed Gondwana indicates the likelihood of a major ocean basin at this time. The final convergence between these blocks is indicated by the clustering of middle Cambrian paleomagnetic poles from cratons of both sides, including extensive Cambrian-aged remagnetizations.

Fig. 10 illustrates a paleomagnetically controlled paleogeographic reconstruction of Laurentia, Baltica and Rio de la Plata–Congo-São Francisco for around 575 Ma. Rio de la Plata and Congo cratons have been positioned according to the new SAn pole, Laurentia according to the 577 Ma Callander Complex paleomagnetic pole (Symons and Chiasson, 1991), and Baltica following Cawood and Pisarevsky (2006). Amazonia and West Africa are reconstructed as still attached to Laurentia, according to matching of the ca. 615 Ma Long Range Dykes (Hodych et al., 2004) and the coeval Adma Diorite (Morel, 1981) poles (see insight Fig. 10). A large ocean between Rio de la Plata–Congo-São Francisco and Amazonia-West Africa, i.e. the Clymene Ocean, is evident in this paleoreconstruction in the middle Ediacaran. Separation of Amazonia–West Africa from Laurentia is very loosely controlled by paleomagnetic data between around 615 and 525 Ma. Major geological evidence indicates that final opening of the eastern passive margin of Laurentia occurred in the latest Ediacaran (565–550 Ma, Cawood et al., 2001). Traditionally, this has been interpreted as final break-up of Rodinia by drifting apart of Amazonia–West Africa from Laurentia (e.g. Li et al., 2008). In this scenario, the existence of a large Clymene ocean as depicted in Fig. 10 is unavoidable. An alternative possibility as proposed by McCausland et al. (2011), suggests an older age for separation of Amazonia (perhaps around 600 Ma) while generation of the eastern passive margin of Laurentia (ca. 550 Ma) is attributed to rifting of peri-Amazonian blocks (i.e. Arequipa-Antofalla and other Proterozoic Andean terranes). However, the Campo Alegre pole

and the Playa Hermosa and Villa Mónica mean virtual geomagnetic poles indicate a large paleolatitudinal gap between Amazonia and Rio de la Plata cratons by ca. 600 Ma. Only if the latter three poles are ruled out and the older age for separation of Amazonia from Laurentia (ca. 600 Ma) is assumed, a large Clymene ocean can be avoided. Reliable paleomagnetic data from the Amazonia and West Africa cratons for the 550–600 Ma lapse is necessary for a definitive answer.

6.2. Implications for Gaskiers-aged glaciation

The position of the new paleomagnetic pole for the Playa Hermosa Formation is not significantly different from the original virtual geomagnetic pole albeit with new constraints of ca. 594–ca. 578 Ma for its age. Although no field test is available for the origin of the characteristic remanence of these sedimentary rocks, a primary remanence is consistent with the pole position close to the 595 Ma Campo Alegre lavas paleomagnetic pole (D'Agrella-Filho and Pacca, 1988). However, the validity of the latter as representative of the Rio de la Plata craton for the Ediacaran has been questioned (Font et al., 2011). Nevertheless, similar positions with the Villa Mónica Fm mean virtual geomagnetic pole (SBf, Rapalini et al., 2013) reinforces the likelihood of a primary origin. The Playa Hermosa pole positions, computed both in situ and after bedding correction of the remanence directions, does not coincide with any reference paleomagnetic pole for South America since the Late Carboniferous (Besse and Courtillot, 2002; Torsvik et al., 2008) suggesting that a remagnetization in the last 320 Ma is unlikely. Magnetic fabric results point to lack of significant inclination shallowing in the Playa Hermosa sediments, which suggest that the 13.0° + 9.5°/−5.5° paleolatitude for deposition of the glacially influenced sediments (inferred from its paleomagnetic pole) favors low latitude glaciation at the Rio de la Plata craton during the Early Ediacaran. However, a field test on the primary nature of this magnetization is needed before more robust conclusions are extracted from these data.

The ca. 594–ca. 578 Ma interval of possible deposition of the Playa Hermosa Formation was contemporaneous with the well-dated accumulation of the Camaquã Supergroup in Brazil (Janikian et al., 2008, 2012). The glacial characteristics observed in the Playa Hermosa Formation contrasts with the absence of glacial characteristics in that basin (Janikian et al., 2008) Taken together with the low latitude of these deposits, these results provide evidence for a low-latitude Gaskiers-aged glaciation in the Rio de la Plata craton in contradiction of the conclusions of Janikian et al. (2008).

6.3. Implications for the Ediacaran geomagnetic field

A long lasting debate has taken place in the last two decades regarding the Ediacaran paleomagnetic database of Laurentia (Symons and Chiasson, 1991; Meert et al., 1994; Pisarevsky et al., 2001; McCausland et al., 2007, and many others). The controversy revolves around highly disparate, high and low inclination magnetic remanences preserved in cratonic rocks of similar age. For example, reliable paleomagnetic poles suggest that Laurentia was located at a low paleolatitude at around 615 Ma (Long Range dykes, Murthy et al., 1992) and again sometime after ca. 560 Ma (e.g., Sept Iles intrusion, Tanczyk et al., 1987; Buckingham flows, Dankers and Lapointe, 1981; Skinner Cove Formation poles, McCausland and Hodych, 1998). However, for a relatively brief period between 590 and 570 Ma, Laurentia appears to have occupied high, nearly polar, latitudes (e.g., Callander Complex, Symons and Chiasson, 1991; Catoctin volcanics, Meert et al., 1994; see also McCausland et al., 2007, and references therein and Table 3). Furthermore, some of these units (Sept-Iles, Catoctin) seem to carry both high and low inclination directions as apparent primary magnetizations. Recent paleomagnetic results from the Baies de Moutons syenite in Eastern Laurentia (McCausland et al., 2011) of ca. 585 Ma, have confirmed the presence of a high latitude component (A) with a second, unrelated, component (B) from slightly younger intrusions of the same complex yielding low paleolatitudes for Laurentia. Abrajevitch and Van der Voo (2010) point out a similar paradox in the Baltica paleomagnetic database, in which the ca. 583 Ma Fen Carbonatite Complex (Meert et al., 1998) and one component of the 584 Ma Alno Carbonatite Complex (Meert et al., 2007) show steep inclinations that would indicate high paleolatitudes for Baltica. Low inclination results from the Alno Carbonatite and the ca. 580 Ma Ukranian traps directions (Elming et al., 2007) suggest low paleolatitudes.

Different explanations have been proposed for this high/low inclination paradox. They range from very fast plate tectonic displacements (Meert et al., 1993, 1994; McCausland et al., 2007), paleomagnetic artifacts due to unreliable data (Pisarevsky et al., 2001, 2008), inertial interchange true polar wander (Evans, 2003; Kirschvink et al., 2005) or unusual geomagnetic geometries (Abrajevitch and Van der Voo, 2010). These last authors ruled out plate tectonic displacements or true polar wander on the basis of non-actualistic or impossibly fast plate motions or pan-lithospheric shifts. Instead, Abrajevitch and Van der Voo (2010) speculate that the middle Ediacaran Earth Magnetic Field had two apparently stable configurations, a geocentric axial dipole (GAD) and a geocentric equatorial dipole (GED). There is no evidence for such non-uniformitarian behavior of the geomagnetic field in Phanerozoic times, and the mechanisms underlying the GED/GAD duality are obscure. However, we can test whether the SAn pole, from rocks now dated at ca. 578 Ma, is consistent with a primary magnetization obtained during this putative GED/GAD behavior of the geomagnetic field.

Like the "high" and "low" latitude poles of similar age from Laurentia and Baltica, the angular distance between SAn and SA1 poles is over 70°. A geomagnetic field changing between a GAD and a GED geometry allows for paleolongitudinal constraints in paleogeographic reconstructions, provided either a single GED state prevailed with a preferred geographical position, as proposed by Abrajevitch and Van der Voo (2010). Fig. 11 shows a paleomagnetically controlled paleogeographic reconstruction of Laurentia, Baltica and Rio de la Plata–Congo-São Francisco cratons for the middle to late Ediacaran (ca. 585–555 Ma). Laurentia and Baltica have been reconstructed assuming the low-latitude option, using the Sept-Iles pole (ca. 564 Ma) and the mean Baltica poles (ca. 555 Ma, Meert et al., 2007), respectively. Rio de la Plata and Congo-São Francisco are considered as already attached or very close to each other and are positioned according to SAn (ca. 578 Ma) obtained in this study. In this reconstruction, the Catoctin volcanic and Callander Complex poles from Laurentia as well as the Fen Complex and the Anlo Carbonatite poles from Baltica fall relatively close to each other, and near the equator, illustrating the hypothesis of the GED configuration. However, the SA1 pole falls quite far from the other group. The relative longitudinal position of the Rio de la Plata–Congo-São Francisco has been selected following Rapalini et al. (2013). There is a clear mismatch between the SA1 pole and the remaining equatorial poles. Overlapping them is not possible since this would require overlapping the continental crusts of Río de la Plata–Congo-São Francisco with that of Baltica. On the other hand, choosing the antipodal location of SA1 and therefore the Rio de la Plata–Congo-São Francisco land masses would place these cratons on an opposite hemisphere to Laurentia, turning highly unlikely the 550–525 Ma accretion of Amazonia and other blocks to form Gondwana (Trindade et al., 2006; Tohver et al., 2012). In summary, our analysis confirms that the SA1 pole is likely a Cambrian

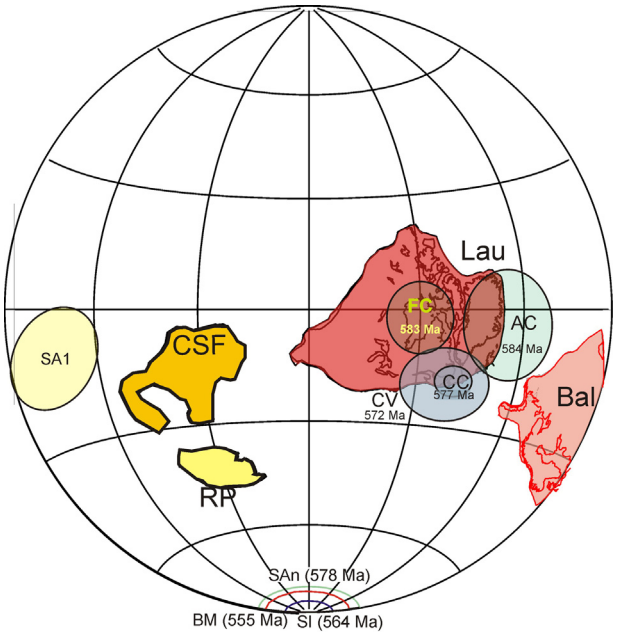


Fig. 11. Mid-Ediacaran paleogeographic reconstruction of Laurentia, Baltica, Rio de la Plata and Congo-São Francisco cratons adopting the "low-latitude" option for the first two. Laurentia is located according to the Sept-Iles pole (SI, 564 Ma), Baltica according to a mean pole of ca. 555 Ma (BM) and Rio de la Plata and Congo-São Francisco according to the new Sierra de las Animas pole (SAn). The geocentric equatorial dipole (GED) hypothesis of Abrajevitch and Van der Voo (2010) is tested by plotting the "high latitude" poles of Laurentia (CV: Catoctin Volcanics, CC: Callander Complex) and Baltica (AC: Alno Carbonatite, FC: Fen Complex) and pole SA1 for the Rio de la Plata craton. See discussion in the text.

remagnetization and indicates that the GAD/GED hypothesis remains to be proved.

7. Conclusions

A new paleomagnetic and geochronologic study on the Sierra de las Ánimas complex and the Playa Hermosa Formation exposed in southern Uruguay yielded significant results. U/Pb (SHRIMP) dating on magmatic zircons allow us to determine that the complex was erupted around 578 Ma and did not extend into the Cambrian. Similar dating on detrital zircons from the Playa Hermosa sediments constrain their deposition to the 595–578 Ma interval, which suggest that they are related to the Gaskiers glaciation. Paleomagnetic analysis permitted to define a more robust and well dated paleomagnetic pole at 578 Ma for the Río de la Plata craton. This new pole (SAn) supersedes the SA2 pole of Sánchez Bettucci and Rapalini (2002). No remanence directions compatible with the SA1 pole determined by these authors, and considered of Cambrian age, were found in the new study. The new geochronologic constraints indicate that rocks from which SA1 was computed are also of middle Ediacaran time. Paleogeographic reconstructions for the middle Ediacaran (ca. 575 Ma) support the existence of a wide Clymene Ocean between the Río de la Plata and Congo-São Francisco cratons on one side and Amazonia and West Africa on the other. New paleomagnetic results on the Playa Hermosa sediments suggest that these glacially related sediments were deposited at low latitudes, although more robust paleomagnetic data is needed. The hypothetical co-existence of two stable configurations of the Earth Magnetic Field, with geomagnetic poles situated either at the geographic poles or at the equator in the middle Ediacaran, cannot explain SA1 pole located near 90° from SAn. A Cambrian remagnetization affecting some units of the Sierra de las Ánimas complex is the simplest explanation for SA1.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.precamres.2014.11.021.

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